



Alberta Public Health Disease Management Guidelines

Flavivirus (Other)



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Health and Wellness Promotion Branch

Public Health and Compliance Branch

Alberta Health

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Case Definition

Probable Case

See *footnote* ^(A)

Laboratory evidence of infection with clinical illness^(B):

- Positive DENV PRNT^(C) and positive ZIKV PRNT

^(A) Due to the cross-reactivity of flaviviruses, dengue serology needs to be interpreted in conjunction in zika serology, when available. This case definition may change if different flaviviruses become more prevalent.

^(B) Clinical illness: depending on the flavivirus involved symptoms can vary from febrile illness with or without rash to encephalitis to hemorrhagic fever with or without hepatitis.

^(C) IgM and/or IgG testing for ZIKV and DENV may be completed as follows: IgM and IgG, IgM only, IgG only). IgM results, if present, indicate likely acute infection however, due to cross-reactivity of flavivirus antibodies, cases with these results (i.e., both DENV and ZIKV PRNT are positive) will be considered probable flaviviruses until further evidence becomes available.

Table 1: Zika and Dengue Lab Result Interpretation⁽¹⁾

(Highlighted light blue are the most important when reviewing results for this case definition)

NOTIFIABLE							
NAT/PCR Testing	Zika Testing ^(D)			Dengue Testing			Interpretation
	ZIKV IgM	ZIKV IgG	ZIKV PRNT	DENV IgM ^(E)	DENV IgG	DENV PRNT	
Negative Not tested	Any result ^(F)	Any result	≥20 (Positive)	Any result	Any result	≥20 (Positive)	PROBABLE FLAVIVIRUS – specific virus cannot be identified due to cross-reacting flavivirus antibodies. Refer to Flavivirus case definition.

Refer to the Alberta Public Health Disease Management Guidelines for [Dengue](#) and [Zika](#) for more information.

^(D) For the purposes of lab result interpretation, a negative PRNT is any value less than 20.

^(E) Individuals tested >3 months after exposure will almost always test IgM negative. If clinically indicated (e.g., pregnancy) samples may be forwarded to NML for PRNT testing with incomplete serology.

^(F) “Any result” includes “Not tested”. Applicable where ever this phrase is used in the table.

Reporting Requirements

Physicians, Health Practitioners and Others

Physicians, health practitioners or others shall notify the Medical Officer of Health (MOH) (or designate) of the zone, of all confirmed and probable cases by mail, fax or electronic transfer within 48 hours (two business days).

Laboratories

All laboratories shall report all positive laboratory results by mail, fax or electronic transfer within 48 hours (two business days) to the MOH (or designate) of the zone and the Chief Medical Officer of Health (CMOH) (or designate).

Alberta Health Services and First Nations Inuit Health Branch (FNIHB)

- The MOH (or designate) of the zone where the case currently resides shall forward the initial Notifiable Disease Report (NDR) of all confirmed and probable cases to the CMOH (or designate) within two weeks of notification and the final NDR (amendment) within four weeks of notification.
- For out-of-province and out-of-country reports, the following information should be forwarded to the CMOH (or designate) by phone, fax or electronic transfer within 48 hours (two business days):
 - name,
 - date of birth,
 - out-of-province health care number,
 - out-of-province address and phone number,
 - positive laboratory report, and
 - other relevant clinical/epidemiological information.

Appendix 1: Revision History

Revision Date	Document Section	Description of Revision
November 2021	General	Updated Template
	Case definition	Updated Laboratory Interpretation Table to be more specific for Flavivirus (Other)

References

1. Rabe IB, Staples JE, Villanueva J, Hummel KB, Johnson JA, Rose L, et al. Interim Guidance for Interpretation of Zika Virus Antibody Test Results. MMWR Morb Mortal Wkly Rep. 2016;65(21):543–6.